

Topology-Aware Immunization of the European Air-Travel Network: Compartmental Models and the Degree–Betweenness Question

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ABSTRACT

The topology of a mobility network decides how an epidemic spreads and which intervention contains it. We model disease propagation on the European subgraph of the OpenFlights air-travel network as a metapopulation reaction–diffusion process, running four compartmental dynamics (SIR, SIS, SEIR, SQIR) with parameters anchored to literature reproduction numbers, and we compare vaccination by random, degree, and betweenness targeting. Our central, testable question is where Europe sits on a known spectrum: in the US domestic network the largest hubs are also the main bridges, so degree- and betweenness-targeted immunization coincide, whereas in the worldwide network geopolitical community structure produces anomalous high-betweenness, low-degree gateways that only betweenness targeting captures. A subgraph- and core-aware layer (k -cores, motifs, graphlets) refines targeting and explains outcomes. The study is a reproducible pipeline whose region and transport layers (air, land, water) are parameters, so the same analysis extends across continents and to multimodal combinations. Claims are comparative, not predictive.

1. INTRODUCTION

A single immunization policy cannot be optimal for every pathogen. We ask how vaccination strategy interacts with both the infection model and the topology of the travel network, on a real, strongly heterogeneous substrate. Concretely, we simulate spread along European aviation routes and test random, degree-, and betweenness-targeted vaccination across four compartmental models, reinforcing the principle that a strategy should be matched to the available infection information rather than applied uniformly. Modelling epidemics on flight networks is not itself new; our specific positioning and the gap we fill are stated in Section 3.

2. RELATED WORK

Compartmental epidemics originate with Kermack and McKendrick [13]; network epidemiology asks how contact structure changes those dynamics [20; 12]. Scale-free topology can destroy the epidemic threshold [21], and Newman linked SIR to bond percolation for exact outbreak results [18]. Because hubs dominate spreading, targeted immunization far outperforms random [22; 4], though the best spread-

ers sit in the network core rather than simply at the highest degree [14]. At the inter-city scale, the air network governs global spread [5; 1], formalised as a metapopulation reaction–diffusion process with a global invasion threshold [6; 7]. Structurally, motifs [17] and graphlet signatures [23] fingerprint local topology but are rarely used to steer immunization — the gap we target.

3. NOVELTY AND POSITIONING

(1) Europe, four models, one protocol. Prior work targets the worldwide or US networks; a Europe-focused comparison across SIR, SIS, SEIR and SQIR under a common protocol is, to our knowledge, missing.

(2) The degree–betweenness question. In hub-and-bridge networks such as the US domestic system, degree and betweenness are highly correlated and the two immunization strategies give nearly identical results [28]. On the worldwide network, Guimerà et al. [9] found anomalous centrality: community and geopolitical structure give some low-degree airports very high betweenness (Anchorage being canonical). Where a network sits on this spectrum decides whether the strategies converge or diverge. We test empirically where Europe lies. Because region is a pipeline parameter (Section 7), we place Europe on the spectrum directly by repeating the analysis across continents and comparing the degree–betweenness correlation.

(3) Subgraph- and core-aware layer. We use k -core decomposition and small-subgraph statistics both to target immunization and to explain why a model–strategy pair behaves as it does, connecting the result to the local structure that produces it.

4. DATA AND NETWORK

The primary substrate is the air layer, from the OpenFlights route database [19], built with NetworkX [10]: nodes are airports, edges are routes weighted by flight frequency. The region is a parameter of network generation (Section 7), so the European subgraph is the focus while other continents and the world follow for the cross-region comparison.

Multimodal data sources. The same node set admits ground and sea layers. For land, rail and road topology is globally available from OpenStreetMap and the GRIP roads database [16]. To keep the cross-region comparison valid, flow weights must be built the same way for every region; we therefore generate them with a single gravity/radiation model calibrated on population [26], applied uniformly, and

use Eurostat commuting data only to validate that model within Europe — never as the production source for one region while others differ. Commuting dominates short-range flux and the proxy choice materially affects predictions [1; 29]. For water, we use OpenStreetMap ferry routes and open cargo port-call data, whose network structure and disease relevance are documented in [11; 25; 15]. Air is in use; land and water enter as additional tagged layers.

Modelling assumptions. (i) Population scales with degree,

$$\text{pop}(v) = P_0 + d(v) P_{\text{route}}, \quad (1)$$

with $P_0 = 150,000$ and $P_{\text{route}} = 45,000$, assuming more routes imply a larger city — a proxy, not census ground truth. (ii) Interventions are biological shields only: vaccinated cities become immune dead-ends and no nodes or edges are removed, isolating the effect of vaccination from route closures. (iii) The travel rate τ is scaled by route frequency, so busier corridors carry proportionally more travellers. (iv) The outbreak starts from a single random patient-zero seed, averaged over seeds. (v) The network is static (no timetables or seasonality). The resulting European grid is a hybridised scale-free network with extreme degree heterogeneity, in which hubs (London, Paris, Frankfurt, Amsterdam) connect high-volume corridors.

5. MODELS AND PARAMETERS

5.1 Metapopulation dynamics with two mobility mechanisms

Each city carries a full compartmental model [6]; the four models span permanent immunity (SIR: measles), no immunity (SIS: common cold), latency (SEIR: COVID-19, SARS) and isolation (SQIR: Ebola). Crucially, mobility is not all of one kind, and treating it as such is a known error [3]. We therefore separate two mechanisms:

Diffusive mobility (air, water). A fraction of every compartment relocates along outbound edges each day and mixes at the destination, split by edge weight and a per-layer travel rate. This is the standard reaction–diffusion coupling and carries a global invasion threshold [7].

Recurrent mobility (land/commuting). Commuters do not relocate: they travel to a neighbour by day, mix there, and return home at night [3; 27]. We implement this as a coupling of the force of infection rather than a relocation: the pressure on city i ’s residents is $\pi_i = \sum_j C_{ij} I_j^*/N_j^*$, where C is the per-capita commuting matrix (the land radiation kernel with a stay-home diagonal) and $I_j^* = \sum_i C_{ij} I_i$, $N_j^* = \sum_i C_{ij} N_i$ are the infectious and total people present at j . With no land layer C is the identity and $\pi_i = I_i/N_i$, recovering the plain diffusive model.

This distinction is not cosmetic. On the European air+land network, modelling commuting as recurrent rather than diffusive lowers the peak active infection by roughly $5\times$ (from $\sim 277\text{M}$ to $\sim 49\text{M}$), because recurrent commuting couples neighbours without permanently transporting the seed population across the continent — exactly the threshold-shifting effect reported by [3].

In-transit transmission (refinement axis). Optionally, travellers can be infected during the journey, which matters most for long confined trips such as ferries [24]. We model this by running the full compartmental reaction on each travelling cohort for the trip duration (distance/speed) at

Table 1: Per-model parameters used in the experiments. Rates are per day; $R_0^{\text{loc}} = \beta/\gamma$ is the well-mixed local reproduction number (the network value is larger, scaled by $\langle k^2 \rangle / \langle k \rangle$).

	SIR	SIS	SEIR	SQIR
Transmission β	0.32	0.32	0.32	0.32
Recovery γ	0.12	0.12	0.12	0.12
Incubation σ	—	—	0.20	—
Quarantine κ	—	—	—	0.15
R_0^{loc}	2.7	2.7	2.7	2.7

an onboard contact rate — so passengers can be infected, recover, incubate or be quarantined en route, not merely toggled infected. An onboard-control parameter (0–1) reduces the onboard transmission, representing screening, onboard quarantine or a hospital ship, so the same machinery answers “how well do onboard measures work?”. This is a toggleable refinement: with it off, transmission occurs only at cities.

5.2 Parameter choice

We fix clinical rates (γ, σ, κ) from each disease’s epidemiology and set β to a literature R_0 . On a heterogeneous network the well-mixed identity $R_0 = \beta/\gamma$ fails: degree fluctuations inflate it as $R_0 \propto \frac{\beta}{\gamma} \frac{\langle k^2 \rangle}{\langle k \rangle}$ [21], and for multi-compartment models we compute R_0 via the next-generation matrix [30].

5.3 Calibration vs. verification

We do not calibrate R_0 to case data as predictive frameworks do [2], because the OpenFlights substrate carries no ground-truth outbreak to fit. Instead we (i) select an operating point in the R_0 regime where strategies actually differ, relative to the threshold [7]; (ii) run a sensitivity sweep showing the ranking of strategies is stable across β , τ , coverage and efficacy; and (iii) verify the implementation against analytic limits (single-node SIR; bond-percolation final size [18]). This recovers calibration’s credibility without fitting, and keeps our claims comparative rather than predictive of absolute case counts.

6. VACCINATION STRATEGIES

With a fixed budget of 15 cities at 75% coverage and 85% efficacy, we compare: control (none); random; degree (densest traffic); and betweenness (structural bridges). We also contrast low (1%) versus high (75%) coverage. As a structural refinement we add a subgraph/core strategy: prioritise innermost k -shells [14] and cities embedded in dense transmission motifs [17; 23] rather than isolated high-degree nodes; the same statistics explain why an intervention dismantles the core or merely trims its periphery.

Expected mechanisms (hypotheses). The same strategy should act differently per model, which is the point of testing all four. Targeting high-betweenness hubs severs Europe’s structural bridges [22], and we hypothesise: for SIR (permanent immunity) this boxes the outbreak into its origin region, where it starves as local susceptibles deplete — a firebreak; for SIS (no immunity), which cannot be eradicated and reaches an endemic plateau, hub immunization acts instead as a continuous filter that compresses the active baseline; for

Table 2: Cross-region degree-betweenness correlation (air layer).

Region	$\rho(\text{deg, btw})$	anomalous gateways
Europe	0.90	9
Asia	0.85	16
Africa	0.78	7
Americas	0.73	59
Oceania	0.67	16

SEIR (latency), where pre-symptomatic passengers defeat departure screening, pre-vaccinating hubs intercepts them at layover nodes; and for SQIR (isolation), shifting identified hub cases into quarantine cuts the busiest corridors. These are hypotheses to be tested, not results.

7. PIPELINE AND SUBSTRATE

The study is a reproducible three-stage pipeline — data retrieval, network generation, evaluation — in which a run is a pure function of its configuration and seed, and the sweep (regions $\times$ layer combinations $\times$ models $\times$ strategies $\times$ coverage $\times$ seeds) is orchestrated with Nextflow [8]. Because the reaction step is substrate-agnostic, network generation emits every combination of the air, land and water layers of Section 4, making multimodality a comparable axis. Layers are kept tagged with their own travel rates, since coupling can cross the epidemic threshold even when no single layer would [7].

8. RESULTS

We build the European air network (561 airports, 10,072 directed routes; $\langle k \rangle = 35.9$, $\langle k^2 \rangle / \langle k \rangle = 117$, confirming strong degree heterogeneity). The simulation horizon is set to 210 days from the operating-point scan rather than by convention: at $\tau = 0.0002$ the latest-peaking model (SEIR) peaks at day ≈ 169 and SIR at day ≈ 90 , so shorter horizons truncate every model before its peak.

The degree-betweenness question. Europe sits at the correlated end of the centrality spectrum, with Spearman $\rho(\text{degree, betweenness}) = 0.90$ and only 9 anomalous low-degree/high-betweenness gateways (e.g. Kittilä, Isles of Scilly, Ivalo). Running the identical construction across regions confirms a true spectrum (Table 2): Europe and Asia are the most US-like, while the Americas ($\rho = 0.73$, 59 gateways) and Oceania ($\rho = 0.67$) are the most worldwide-like (anomalous). Europe is therefore correlated but not extreme — it carries its own peripheral gateways.

Vaccination strategies. Across all four models, targeted allocation outperforms random, which barely beats no intervention. For SIR at high coverage the mean reduction in peak active infections relative to control is: degree -11.4% , subgraph -11.2% , betweenness -9.0% , k -core -8.4% , random -1.4% . That degree edges out betweenness is consistent with Europe’s high ρ : where the two centralities largely agree, both target the same hubs, so the choice matters less — precisely the regime-dependence the paper set out to test.

9. CONCLUSIONS

We posed a Europe-specific, multi-model test of whether immunization strategy must be matched to network topology, centred on the degree-betweenness question. Empir-

ically Europe is degree-betweenness correlated ($\rho = 0.90$), so degree- and betweenness-targeting nearly coincide and both beat random allocation by roughly 9–11% in peak suppression; the divergence the literature reports for the worldwide network appears, in our cross-region comparison, in the Americas and Oceania rather than in Europe. Limitation: populations and the travel rate are synthetic proxies, so these are comparative, not predictive, results; the land/water layers currently use radiation-model flows rather than measured rail/road geometry.

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